AE1MCS: Mathematics for Computer Scientists

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Reading

Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 7th Edition, 2013.

- Chapter 7, Section 7.1 An Introduction to Discrete Probability
- Chapter 7, Section 7.2 Probability Theory
- Chapter 7, Section 7.3 Bayes' Theorem
- Chapter 7, Section 7.4 Expected Value and Variance

Discrete Probability

- Combinatorics and probability theory share common origins (analyzing gambling games).
- The theory of probability now plays an essential role in a wide variety of disciplines (e.g. the study of genetics).
- In computer science,
  - Probability theory plays an important role in the study of the complexity of algorithms.
  - Probabilistic algorithms vs. deterministic algorithms.
  - Probability theory can help us answer questions that involve uncertainty.
  - ...

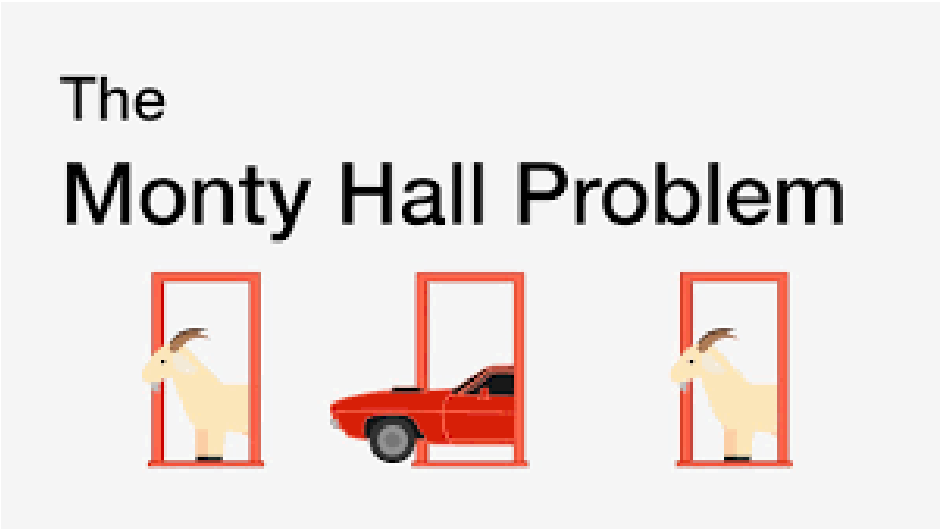
Content

- Probability of an Event
- Probabilities of Complements and Unions of Events

Monty Hall Three-Door Puzzle

You are asked to select one of three doors to open; the large prize is behind one of the three doors and the other two doors are losers. Once you select a door, the game show host, who knows what is be behind each door:

- whether or not you selected the winning door, he opens one of the other two doors that he knows is a losing door (selecting at random if both are losing doors).
- Then he asks you whether you would like to switch doors.



Finite Probability

Laplace's definition of the probability of an event with **finitely many, equally likely, possible outcomes** is as follows.

Definition

If S is a finite nonempty sample space of equally likely outcomes, and E is an event, that is, a subset of S, then the probability of E is

$$p(E) = \frac{|E|}{|S|}$$

- An **experiment** is a procedure that yields one of a given set of possible outcomes.
- The **sample space** of the experiment is the set of possible outcomes.
- An **event** is a subset of the sample space.

Probability Distribution

Let S be the sample space of an experiment with a finite or countable number of outcomes. We assign a probability $p(s)$ to each outcome $s \in S$. We require that two conditions be met:

1. $0 \leq p(s) \leq 1$ for each $s \in S$
2. $\sum_{s \in S} p(s) = 1$.

The function p from the set of all outcomes of the sample space S is called a **probability distribution**.

Probability of an Event

In the eighteenth century, the French mathematician Laplace, who also studied gambling, defined **the probability of an event as the number of successful outcomes divided by the number of possible outcomes**.

Example: Poker 1

Find the probability that a hand of five cards in poker contains four cards of one kind.

- A deck of cards contains 52 cards.
- There are 13 different kinds of cards, with four cards of each kind.
- These kinds are twos, threes, fours, fives, sixes, sevens, eights, nines, tens, jacks, queens, kings, and aces.
- There are 4 suits: spades, clubs, hearts, and diamonds, each containing 13 cards.

Example: Poker 1 (Answer)

Find the probability that a hand of five cards in poker contains four cards of one kind.

- S is number of ways to choose any 5 cards from 52: $|S| = C(52, 5)$.
- E is the number of ways to get four of a kind: $|E| = 13 \times (52 - 4)$, using product rule with
  - Choose one of the 13 kinds which is repeated 4 times;
  - Choose any remaining card for last card (52-4).
- Hence $p(E) = \dfrac{13 \times (52 - 4)}{C(52, 5)} \approx 0.00024$.

Example: Poker 2

What is the probability that a poker hand contains a full house, that is, three of one kind and two of another kind?

Example: Poker 2 (Answer)

What is the probability that a poker hand contains a full house, that is, three of one kind and two of another kind?

Same sample space: $|S| = C(52, 5)$.

$$
\begin{array}{rcll}
|E| & = & \text{\# ways to get two different kinds} & C(13,2) \\
 & \times & \text{\# ways to select which is three cards of the same kind} & 2 \\
 & \times & \text{\# ways to choose three cards of the same kind} & C(4,3) \\
 & \times & \text{\# ways to choose two cards of the same kind} & C(4,2) \\
 & = & C(13,2) \times 2 \times C(4,3) \times C(4,2) &
\end{array}
$$

Notice: $C(13,2) \times 2 = P(13,2) = C(13,1)C(12,1)$*, so all correct.*

Probabilities of Complements and Unions of Events

Theorem

Let E be an event in a sample space S. The probability of the event $\overline{E} = S - E$, the complementary event of E, is given by

$$p(\overline{E}) = 1 - p(E)$$

Theorem

Let E_1 and E_2 be events in the sample space S. Then

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

How to prove them?

Example

- A sequence of 10 bits is randomly generated. What is the probability that at least one of these bits is 0?
- What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?

Example 1 (Answer)

A sequence of 10 bits is randomly generated. What is the probability that at least one of these bits is 0?

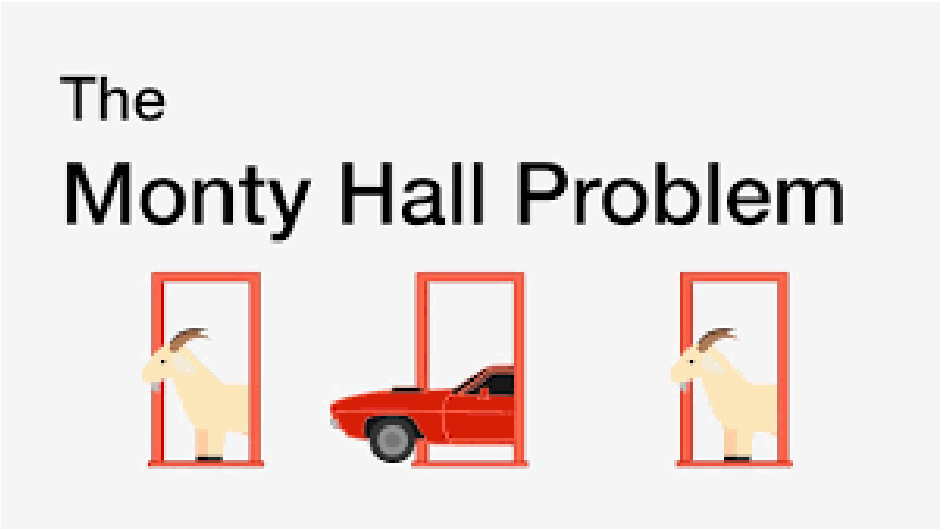
- Sample space is all bit strings of length 10: $|S| = 2^{10}$.
- Think about the event when bit string has *no* 0's. Then $E = \{1111111111\}$ and $|E| = 1$.
- Now the event we are interested in (at least one 0) is complement of E, so

$$p(\overline{E}) = 1 - p(E) = 1 - 2^{-10}.$$

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Monty Hall Three-Door Puzzle

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1. $0 \leq p(s) \leq 1$ for each $s \in S$
2. $\sum_{s \in S} p(s) = 1$.

The function p from the set of all outcomes of the sample space S is called a **probability distribution**.

Conditional Probability

Given an event F occurs, the probability that event E occurs is the **conditional probability** of E given F.

Let E and F be events with $p(F) > 0$. The **conditional probability** of E given F, denoted by $p(E|F)$, is defined as:

$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

.

Conditional Probability

A bit string of length four is generated at random so that each of the 16 bit strings of length four is equally likely. What is the probability that it contains at least two consecutive 0s, given that its first bit is a 0?

Solution: Let E be the event that a bit of length four contains at least two consecutive 0s,
Let F be the event that the first bit of a bit string of length four is a 0.
The probability that a bit string of length four has at least two consecutive 0s, given that its first bit is a 0, equals:

$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

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$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

Because $E \cap F = \{0000, 0001, 0010, 0011, 0100\}$, then $p(E \cap F) = \frac{5}{16}$. Because there are 8 bit strings of length four that start with a 0, we have $p(F) = \frac{8}{16} = \frac{1}{2}$.

$$p(E|F) = \frac{5/16}{1/2} = \frac{5}{8}$$

Independence

When two events are independent, the occurrence of one of the events gives no information about the probability of that the other event occurs.
The events E and F are independent **if and only if** $p(E \cap F) = p(E)p(F)$.

Independence

Suppose E is the event that a randomly generated bit string of length four begin with a 1 and F is the event that this bit string contains an even number of 1s. Are E and F independent, if the 16 bit strings of length four are equally likely?

Solution: There are eight bit strings of length four that begin with a one: 1000, 1001, 1010, 1011, 1100, 1101, 1110, and 1111. There are also eight bit strings of length four that contain an even number of ones: 0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111. Because there are 16 bit strings of length four, it follows that

$$p(E) = p(F) = 8/16 = 1/2.$$

Because $E \cap F = 1111, 1100, 1010, 1001$, we see that $p(E \cap F) = 4/16 = 1/4$.
Because $p(E \cap F) = 1/4 = (1/2)(1/2) = p(E)p(F)$,we conclude that E and F are independent.

Bayes' Theorem

Suppose we know p(F), the probability that an event F occurs, but we have knowledge that an event E occurs.
The conditional probability that F occurs given that E occurs, $p(F|E)$

Bayes' Theorem

Bayes' Theorem

Suppose that E and F are events from a sample space S such that $P(E) \neq 0$ and $P(F) \neq 0$. Then

$$p(F|E) = \frac{p(E|F)p(F)}{p(E|F)p(F) + p(E|\overline{F})p(\overline{F})}$$

Bayesian Spam Filters

Suppose that we have found that the word ”Rolex” occurs in 250 of 2000 messages known to be spam and in 5 of 1000 messages known not to be spam. Estimate the probability that an incoming message containing the word “Rolex” is spam, assuming that it is equally likely that an incoming message is spam or not spam. If our threshold for rejecting a message as spam is 0.9, will we reject such messages?

Solution: We use the counts that the word “Rolex” appears in spam messages and messages that are not spam to find that p(Rolex) = 250/2000 = 0.125 and q(Rolex) = 5/1000 = 0.005.
Because we are assuming that it is equally likely for an incoming message to be spam as it is not to be spam, we can estimate the probability that an incoming message containing the word “Rolex” is spam by

$$r(Rolex) = \frac{p(Rolex)}{p(Rolex) + q(Rolex)} = \frac{0.125}{0.125 + 0.005} = 0.962$$

Generalizing Bayes' Theorem

Suppose that E is an event from a sample space S and that $F_1, F_2, ..., F_n$ are mutually exclusive events such that $\cup_{i=1}^{n} F_i = S$. Assume that $p(E) \neq 0$ and $p(F_i) \neq 0$ for $i = 1, 2, ..., n$. Then

$$P(E) = \sum_{i=1}^{n} p(E|Fi)p(Fi)$$

$$p(F_j|E) = \frac{p(E|F_j)p(F_j)}{\sum_{i=1}^{n} p(E|Fi)p(Fi).}$$

Expected Value and Variance

The **expected value**, also called the expectation or mean, of the random variable X on the sample space S:

$$E(X) = \sum_{s \in S} p(s)X(S)$$

If X is a random variable and $p(X = r)$ is the probability that $X = r$, so that $p(X = r) = \sum_{s \in S, X(s)=r} p(s)$, then

$$E(X) = \sum_{r \in X(S)} p(X = r)r$$

Linearity of Expectations

If X_i, $i = 1, 2, ..., n$ with n a positive integer, are random variables on S, and if a and b are real numbers, then

1. $E(X_1 + X_2 + \cdots + Xn) = E(X_1) + E(X_2) + \cdots + E(X_n)$
2. $E(aX + b) = aE(X) + b$.

Independent Random Variables

The random variables X and Y on a sample space S are independent if

$$p(X = r1 \text{ and } Y = r2) = p(X = r1) \cdot p(Y = r2)$$

or, if the probability that $X = r1$ and $Y = r2$ equals the product of the probabilities that $X = r1 and Y = r2$, for all real numbers $r1$ and $r2$.

Corollary

If X is independent of Y, then

$$E(XY) = E(X) \cdot E(Y) \tag{1}$$

If $X_1, X_2, \ldots, X_n$ are mutually independent, then,

$$E(X_1 X_2 \ldots X_n) = E(X_1)E(X_2) \ldots E(X_n) \tag{2}$$

Bernoulli Trials and Binomial Distribution

Each performance of an experiment with two possible outcomes is called a **Bernoulli trial**.
In general, a possible outcome of a Bernoulli trial is called a **success** or a **failure**.

- Generate a bit, $\{0, 1\}$.
- Flip a coin, $\{\text{Heads, tails}\}$.

If p is the probability of a success and q is the probability of a failure, it follows that $p + q = 1$.

Binomial Distribution

- The probability of exactly k successes in n independent Bernoulli trials, with probability of success p and probability of failure $q = 1 - p$, is $C(n, k)p^k q^{n-k}$.
- We denote by $b(k; n, p)$ the probability of k successes in n independent Bernoulli trials with probability of success p and probability of failure $q = 1 - p$.
- Considered as a function of k, we call this function the **binomial distribution**.

$$b(k; n, p) = C(n, k)p^k q^{n-k}.$$

Binomial Distribution

Example

A coin is biased so that the probability of heads is 2/3. What is the probability that exactly four heads come up when the coin is flipped seven times, assuming that the flips are independent?

Expected Value: Extra Exercises (Not required!)

Bernoulli trials:
The expected number of successes when n mutually independent Bernoulli trials are performed, where p is the probability of success on each trial, is np.

Proof: Let X be the random variable equal to the number of successes in n trials. By Theorem 2 of Section 7.2 we see that $p(X = k) = C(n, k)p^k q^{n-k}$. Hence, we have

$$
\begin{aligned}
E(X) &= \sum_{k=1}^{n} kp(X = k) && \text{by Theorem 1} \\
&= \sum_{k=1}^{n} kC(n, k)p^k q^{n-k} && \text{by Theorem 2 in Section 7.2} \\
&= \sum_{k=1}^{n} nC(n-1, k-1)p^k q^{n-k} && \text{by Exercise 21 in Section 6.4} \\
&= np \sum_{k=1}^{n} C(n-1, k-1)p^{k-1} q^{n-k} && \text{factoring } np \text{ from each term} \\
&= np \sum_{j=0}^{n-1} C(n-1, j)p^j q^{n-1-j} && \text{shifting index of summation with } j = k-1 \\
&= np(p+q)^{n-1} && \text{by the binomial theorem} \\
&= np. && \text{because } p+q = 1
\end{aligned}
$$

This completes the proof because it shows that the expected number of successes in n mutually

Geometric Distribution

A random variable X has a geometric distribution with parameter p if $p(X = k) = (1 - p)^{k-1}p$ for $k = 1, 2, 3, \ldots$, where p is a real number with $0 \leq p \leq 1$.

$$E(X) = 1/p$$

Geometric Distribution

Suppose that the probability that a coin comes up tails is p. This coin is flipped repeatedly until it comes up tails. What is the expected number of flips until this coin comes up tails?
The random variable X that equals the number of flips expected before a coin comes up tails is an example of a random variable with a geometric distribution.

Expectation

Corollary

If X is a random variable and $P(X = i)$ is the probability that $X = i$, then

$$E(X) = \sum_{i=1}^{\infty} iP(X = i)$$

Theorem

If $X : S \to \mathbb{N}$, then

$$E(X) = \sum_{i=0}^{\infty} P(X > i) = \sum_{i=1}^{\infty} P(X \geq i)$$

System Fail Problem

A system fails with probability p at each step. we assume mutually independent between each step. What is the expected number of steps before the system Fail?

Baby Problem

Some couple wants to get a baby girl. If they get a baby boy, they keep trying until they get a girl. What is the expected number of boys before they get a baby girl?

Linearity of Expectations

If X_i, $i = 1, 2, ..., n$ with n a positive integer, are random variables on S, and if a and b are real numbers, then

1. $E(X_1 + X_2 + \cdots + Xn) = E(X_1) + E(X_2) + \cdots + E(X_n)$
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or, if the probability that $X = r1$ and $Y = r2$ equals the product of the probabilities that $X = r1 and Y = r2$, for all real numbers $r1$ and $r2$.

Corollary

If X is independent of Y, then

$$E(XY) = E(X) \cdot E(Y) \tag{1}$$

If $X_1, X_2, \ldots, X_n$ are mutually independent, then,

$$E(X_1 X_2 \ldots X_n) = E(X_1)E(X_2) \ldots E(X_n) \tag{2}$$

Rolling two dice

Roll two six-sided fair and independent dice. What's the expected product of the dice?

Variance

Betting game.
Betting 10RMB.
Betting 10,000RMB.

Variance

Variance provides a measure of how widely X is distributed about its expected value.

Definition

Let X be a random variable on a sample space S. The variance of X, denoted by $Var(X)$, is

$$Var(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s).$$

That is, Var(X) is the weighted average of the square of the deviation of X. The standard deviation of X, denoted $\sigma(X)$, is defined to be $\sqrt{Var(X)}$.

Variance

Theorem

If X is a random variable on a sample space S, then
$Var(X) = E(X^2) - E(X)^2$

Corollary

If X is a random variable on a sample space S and $E(X) = \mu$, then
$Var(X) = E((X - \mu)^2)$.

How to prove it?

Example: Rolling a Die

Let X be the number that comes up when a fair die is rolled. What is the expected value and variance of X?

Example: Rolling a Die

Let X be the number that comes up when a fair die is rolled. What is the expected value and variance of X?

Solution: The random variable X takes the values $1, 2, 3, 4, 5, 6$, each with probability 1/6. It follows that:

$$E(X) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 = 21/6 = 7/2 = 3.5$$

$$Var(X) = \frac{1}{6} \cdot (1-3.5)^2 + \frac{1}{6} \cdot (2-3.5)^2 + \frac{1}{6} \cdot (3-3.5)^2 + \frac{1}{6} \cdot (4-3.5)^2 + \frac{1}{6} \cdot (5-3.5)^2$$

$$+ \frac{1}{6} \cdot (6-3.5)^2 = 2.917$$

Example

What is the variance of the random variable $X((i,j)) = 2i$, where i is the number appearing on the first die and j is the number appearing on the second die, when two fair dice are rolled?

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What is the variance of the random variable $X((i,j)) = 2i$, where i is the number appearing on the first die and j is the number appearing on the second die, when two fair dice are rolled?

Solution: Note that because $p(X = k)$ is $1/6$ for $k = 2, 4, 6, 8, 10, 12$ and is 0 otherwise,
$E(X) = (2 + 4 + 6 + 8 + 10 + 12)/6 = 7$,
and

$$E(X^2) = (2^2 + 4^2 + 6^2 + 8^2 + 10^2 + 12^2)/6 = 182/3.$$

It follows that:

$$Var(X) = E(X^2) - E(X)^2 = 182/3 - 49 = 35/3.$$

Variance for the sum of random variables

If X and Y are independent variable,

$$Var(X + Y) = Var(X) + Var(Y)$$

In addition,

$$Var(aX + b) = a^2 Var(X)$$

Poisson Distribution

Examples

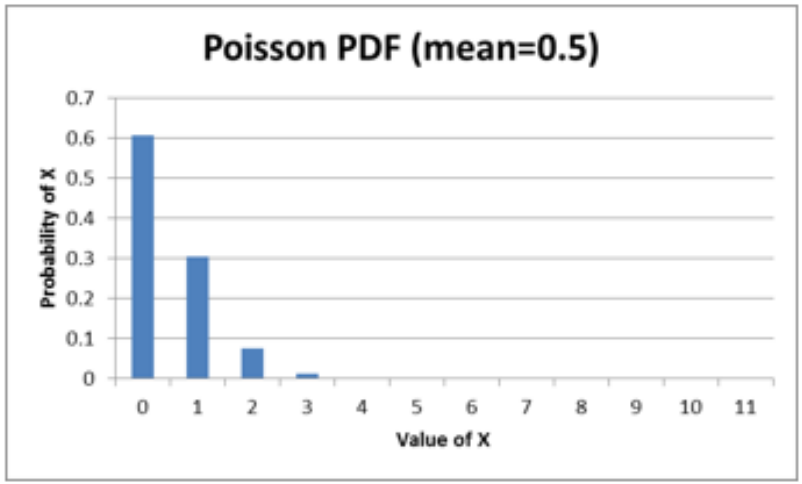
- Process with random arrivals - probability of seeing x events within a certain time period
- Number of slow moving items sold per day, week, month in a store

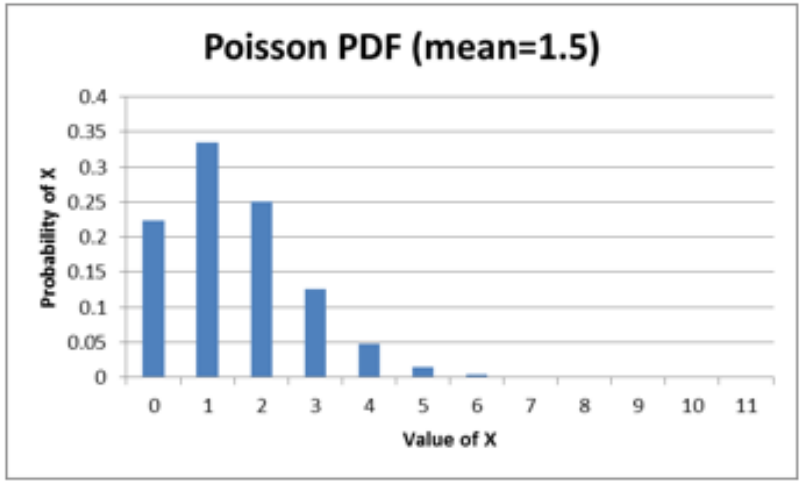
$$P(X = i; \lambda) = \begin{cases} \frac{e^{-\lambda}\lambda^i}{i!}, & \text{for } i = 0, 1, 2, \ldots \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

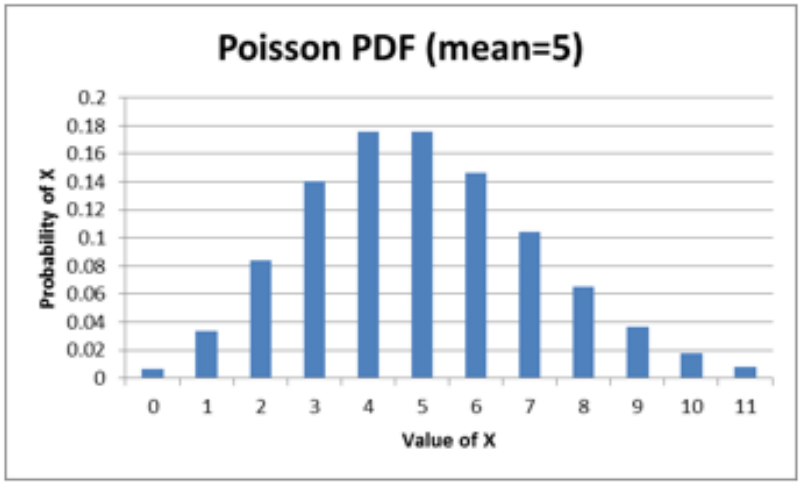
For Poisson Distribution, E(X)=Var(X)=λ

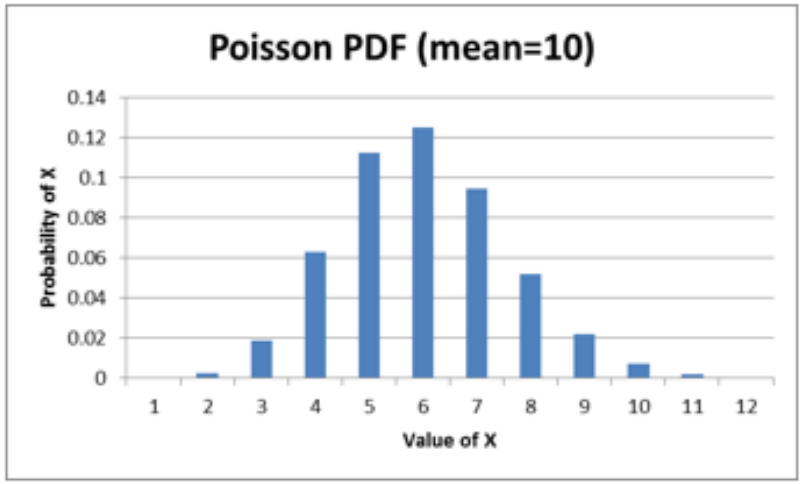
Source:
`https://en.wikipedia.org/wiki/Poisson_distribution`

Poisson Distribution (Not required!)









Example of Poisson Distribution (Not required!)

Suppose that trains arrive at Ningbo Station P(1.5) per 10 minute increments. That is, 1.5 trains on average arrive every 10 minutes and the variance is 1.5 and the distribution appears to be Poisson.

- What is the probability that I will see 3 or more trains in 10 minutes?
- What is the probability none will come by?

Discrete Probability

- Uniform (e.g. rolling a die)
- Bernoulli (Success or Failure)
- Binomial (Number of successes in fixed number of trials)
- Geometric (Number of trials until success)
- Poisson (Number of arrivals in fixed time interval)
- Many others...

Reading

Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 7th Edition, 2013.

- Chapter 7, Section 7.3 Bayes' Theorem
- Chapter 7, Section 7.4 Expected Value and Variance